



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21BT402 / 20BT402 – HEAT AND MASS TRANSFER

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

2) Linear graph paper will be provided.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain different modes of heat transfer. Derive an expression for heat flow through a thick-walled cylinder by conduction. Take r_1 and r_2 as the inner and outer radii of cylinder, k as a mean thermal conductivity. Assume T_1 as the inside temperature and T_2 as the outside temperature.	10	L*3	1	1
	b) With neat, labeled sketch explain the construction and working of plate type heat exchanger. Mentions its application in bio-industries.	10	L2	2	1
2.	a) A 165 mm external diameter pipe is lagged with a 35 mm thickness of lagging of which thermal conductivity is 0.041 W/ (m.K). The lagging outside temperature is 308 K, and brine temperature at a section inside the pipe is 252 K. Find the rise in temperature of the brine per meter length at this section, if the brine mass flow rate is 0.32 kg/s. Specific heat of brine is 3.6 kJ/(kg.K).	10	L3	1	1
	b) The end product in a pharma industry contains 5% solutes, is to be concentrated to a level of 20% solutes (by weight) at a rate of 5000kg/h. Dry saturated steam at a pressure corresponding to saturation temperature of 399K is used. Feed is at 298K, and boiling point elevation (BPE) is 5K. Overall heat transfer coefficient is 2350W/m ² . K. Calculate the economy of the evaporator and area of heat transfer to be provided. Data: Treating solution as pure water and neglecting BPE. The latent heat of condensation of steam at 399K=2185kJ/kg. K. Latent heat of vaporization of water at 1atm and 373K=2257kJ/kg. K. Specific heat of feed=4.187kJ/kg. K.	10	L3	2	2
3.	a) Explain in brief heat transfer to boiling liquid	10	L2	1	1
	b) Antibiotics leaving the reactor are at high temperature and need to cool to room temperature using 1-2 pass shell and tube heat exchanger (STHE). With neat, labeled sketch explain how the temperature is brought down to room temperature in STHE? Also explain the major components of STHE.	10	L2	2	2
Unit – II					
4.	a) Define (i) Molar Concentration (ii) Molar Flux (ii) Mass average velocity (iv) Molecular Diffusion (v) Diffusivity	10	L2	3	2

P.T.O.

21BT402 / 20BT402 Supplementary Examinations - December 2025

- b) The equilibrium adsorption of acetone on an activated charcoal at 303K is given by the following data:

g adsorbed/ g charcoal	0	0.1	0.2	0.3	0.3
Partial Pressure of acetone, mmHg	0	2	12	42	92

The vapor pressure of acetone at 303K is 283mmHg. A 1L flask contains air and acetone vapor at 1atm and 303K, with a relative saturation of vapor at 35%. After 2g of fresh activated charcoal has been added into the flask, flask is sealed. Calculate the final vapor concentration at 303K and the final pressure neglecting the adsorption of air.

10 L3 4 2

5. a) Derive an expression to calculate the diffusion of sterile oxygen through stagnant liquid media inside a stirred tank bioreactor. State the assumptions wherever required.
- b) A Solution containing 5% acetaldehyde and 95% toluene is to be extracted with water in 5 stage cross current extraction unit to extract acetaldehyde. Toluene and water are insoluble. If 25kg of water each time are used per 100kg of feed, calculate the amount of acetaldehyde extracted and final concentration of the exit solution. The equilibrium relation is given as $Y=2.2X$, Where Y is the kg acetaldehyde per kg water, X is the kg acetaldehyde per kg toluene.
6. a) List and explain the theories of mass transfer.
- b) Explain the importance of ternary diagram in extraction process. With the help of ternary diagram explain binodal curve.

10 L3 3 2

10 L3 4 2

10 L2 3 2

10 L2 4 2

Unit – III

7. a) With suitable diagram explain the various plates used for phase contacting in distillation column.
- b) Liquid mixture containing 40mol% methanol and 60mol% water is fed to the differential distillation at atmospheric pressure with 60mol% of the liquid is distilled. Find the composition of the distillate and residue.

Equilibrium data:

x	0.05	0.1	0.2	0.3	0.4	0.5
y	0.27	0.42	0.57	0.66	0.73	0.78

12 L3 5 2

8. a) What is flooding and weeping in distillation column? Explain
- b) A mixture of benzene and toluene containing 60 mol% benzene is to be separated to give a product of 95 mole % benzene and a bottom product containing 10 mol% benzene. The feed enters a column at its bubble point. It is proposed to operate the column with reflux ratio of 2.5. It is required to find the number of theoretical plates needed and the position of feed plate.

5 L2 5 2

Equilibrium Data:

x	0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.13	0.21	0.38	0.5	0.6	0.7	0.77	0.83	0.9	0.95	1.0

15 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
